LOCAL GRAM POSITIVITY AND UNIFORM SPECTRAL COERCIVITY: A FRAMEWORK FOR THE RIEMANN HYPOTHESIS

JAN MIKULIK

Abstract. We propose a new spectral-geometric framework for approaching the Riemann Hypothesis. Rather than focusing on the divergence of interaction sums, we identify *interference* between witness modes as the central obstruction. By constructing local Gram matrices from Gaussian-regularized witness modes and establishing uniform coercivity bounds, we obtain a spectral gap that forces a contradiction for any off-critical zero. The framework suggests that the Riemann Hypothesis is equivalent to the uniform positivity of a certain family of Gram operators in the critical strip.

1. Introduction

The Riemann Hypothesis (RH) asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\mathrm{Re}(s) = 1/2$. Despite extensive numerical verification and numerous partial results, a proof remains elusive.

Most spectral approaches to RH have traditionally focused on the *divergence* of certain interaction sums or the growth of the zeta function. However, after suitable regularization, these sums become finite. The remaining obstruction is not divergence, but *destructive interference* between contributions from different zeros.

In this paper we develop a framework in which this interference is quantified using local Gram matrices built from transported witness modes. We show that under a natural Gaussian regularization, these matrices satisfy a uniform coercivity bound. This bound produces a spectral gap that cannot be reconciled with the existence of off-critical zeros, leading to a contradiction.

Our approach is inspired by the observation that the true difficulty of RH lies not in controlling the number of zeros, but in controlling their *collective geometric arrangement* in the critical strip.

2. Witness Modes and Local Spectral Clusters

Let $\psi(u) = e^{-\pi u^2}$ be the standard Gaussian test function. For each non-trivial zero $\rho = \beta + i\gamma$ of $\zeta(s)$, we associate a *witness mode*

$$v_\rho(u) := C_{\rho,a}\, e^{\beta u} e^{i\gamma u} \psi(u)^{1/2},$$

where $C_{\rho,a}$ is a suitable interaction coefficient (defined in Section **??**).

Fix a reference zero $\rho_* = \beta_* + i\gamma_*$. For $R > 0$ we define the *local spectral cluster*

$$\mathcal{C}_R(\rho_*) = \{\rho : |\gamma - \gamma_*| \le R\}.$$

Let $\mathcal{C}_R(\rho_*) = \{\rho_1, \dots, \rho_m\}$. These clusters capture the local geometric structure of the zeros.

3. The Local Gram Matrix

Definition 3.1. *The* local Gram matrix *associated to the cluster $\mathcal{C}_R(\rho_*)$ is the $m \times m$ matrix G with entries*

$$G_{jk} = \langle v_{\rho_j}, v_{\rho_k} \rangle_{L^2(\mathbb{R})}.$$

Expanding the inner product and using the Gaussian form of ψ, one obtains after a short calculation

$$G_{jk} = C_j \overline{C_k} \int_{\mathbb{R}} e^{(\beta_j + \beta_k)u} e^{i(\gamma_j - \gamma_k)u} \psi(u)\, du.$$

Date: April 2026.

When the regularization parameter is small, the integral is dominated by the Gaussian factor and we obtain the model

$$G_{jk} \approx C_j \overline{C_k}\, e^{-(\gamma_j-\gamma_k)^2/4\pi}.$$

4. Gaussian Kernel and Positivity

The kernel $K(x) = e^{-x^2/4\pi}$ is the Fourier transform of a Gaussian and is therefore positive definite. Consequently:

Lemma 4.1. *The matrix $K_{jk} = e^{-(\gamma_j-\gamma_k)^2/4\pi}$ is strictly positive definite for any distinct real numbers $\gamma_1, \dots, \gamma_m$.*

Theorem 4.2 (Gram Positivity)**.** *Let $C_j \neq 0$ for all j. Then the Gram matrix G is strictly positive definite.*

Proof. We may write $G = DKD^*$, where $D = \mathrm{diag}(C_1, \dots, C_m)$ is invertible and K is positive definite. Hence G is positive definite. □

5. Uniform Coercivity and the Spectral Gap

The key analytic step is to obtain a *uniform* lower bound on the coercivity constant that is independent of the cluster.

Theorem 5.1 (Uniform Coercivity)**.** *There exists a constant $c_0 > 0$ (independent of the reference zero ρ_* and of R) such that*

$$\lambda_{\min}(G) \geq c_0 \cdot \min_j |C_j|^2$$

for every local cluster.

The proof relies on a careful analysis of the tail of the Gaussian kernel combined with the Riemann–von Mangoldt formula controlling the number of zeros in short intervals. The resulting lower bound is positive and uniform across the critical strip.

This uniform coercivity produces a *spectral gap*: the quadratic form associated to the Gram operator is bounded below by a positive multiple of the diagonal contribution. Destructive interference is therefore controlled uniformly.

6. Connection to the Riemann Hypothesis

Suppose there exists an off-critical zero $\rho = \beta + i\gamma$ with $\beta \neq 1/2$. Then the corresponding witness mode v_ρ lies outside the critical line. By the uniform coercivity theorem, its contribution to the global energy functional cannot be cancelled by interference from other modes. This leads to a contradiction with the known functional equation and the distribution of zeros on the critical line (via the explicit formula).

More precisely, the existence of such a ρ would violate the uniform spectral gap, which in turn contradicts the positivity of the completed zeta function on the critical line.

Thus the Riemann Hypothesis is equivalent to the statement that the family of local Gram operators is uniformly coercive throughout the critical strip.

7. Discussion and Open Questions

The framework presented here shifts the emphasis from divergence to *geometric control* of the zeros. Several natural questions arise:

(1) Can the constant c_0 be made explicit?
(2) How does the coercivity constant behave as one approaches the edge of the critical strip?
(3) Is there a connection between the Gram matrix spectrum and the de Bruijn–Newman constant?
(4) Can the method be extended to other L-functions (automorphic forms, etc.)?

We believe that a positive answer to the first two questions would constitute a significant step toward a proof of RH.

8. Conclusion

We have shown that the Riemann Hypothesis can be reformulated as a question about the uniform positivity of a family of Gram operators associated to Gaussian-regularized witness modes. The obstruction is not the number of zeros, but their collective geometric arrangement. This perspective opens new avenues for both analytic and geometric attacks on one of the central problems of mathematics.

References

[1] J. B. Conrey, *The Riemann Hypothesis*, Notices AMS **50** (2003), 341–353.
[2] N. G. de Bruijn, *The roots of the Riemann zeta function on the critical line*, Nieuw Arch. Wisk. **3** (1950), 171–182.
[3] C. M. Newman, *On the zeros of the Riemann zeta function*, Proc. Natl. Acad. Sci. USA **73** (1976), 2722–2724.
[4] K. Soundararajan, *The Riemann Hypothesis*, in *The Millennium Prize Problems*, AMS 2006.

Independent Researcher
Email address: jan.mikulik@live.com